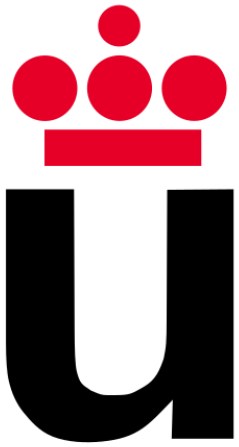
Lab 2: Medical Image Segmentation

Medical Image Analysis
Biomedical Engineering 2026-2027



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AI Disclaimer and Workload Distribution

Sections	Completed by
Image loading, ROI definition and reference segmentations (3D Slicer)	Álvaro Gil Burgo
Thresholding	Álvaro Gil Burgo
Region growing	Álvaro Gil Burgo
Watershed	Álvaro Gil Burgo
Pre- and post-processing	Álvaro Gil Burgo
Active contours (snakes)	César Manuel Robalo Fariñas
Chan-Vese: method and parameter study	César Manuel Robalo Fariñas
Chan-Vese: robustness and final comparison	Álvaro Gil Burgo
Report	Álvaro Gil Burgo and César M. Robalo

Supplemental assistance was provided by Generative AI tools (Claude) under course rules to help develop this laboratory project. AI tools were specifically used to:

- Help to implement, debug and optimise particular code routines (the *src/* modules and the notebook cells), following the steps and decisions set by the group.
- Clarify theoretical concepts related to the segmentation methods.

The regions of interest and the manual reference segmentations in 3D Slicer were made by the group members, all the code was executed and checked by them, and the report was done by both members (each their corresponding part).

Part 1: Thresholding, Region Growing and Watershed

1. Images, ROI Definition and Reference Segmentations

In this practice we used the same two images as in Lab 1: a T1-weighted axial brain MRI (511 × 460 px) and a posteroanterior (PA) chest X-ray (1956 × 2412 px). Both PNG files are stored as RGBA, so they were first converted to a single grey-level channel and normalised to [0, 1].

ROI definition. In each image we chose one anatomical structure and a rectangular region of interest (ROI) around it, given as a bounding box $img[y_{min}:y_{max}, x_{min}:x_{max}]$:

- **Brain MRI:** a focal hypointense lesion in the frontal white matter. ROI = [120:205, 125:215], 85 × 90 px (N_{ROI} = 7,650 px, 3.3 % of the image).
- **Chest X-ray:** the right lung field (left side of the image, following the radiological convention). ROI = [140:1910, 130:1130], 1770 × 1000 px (N_{ROI} = 1,770,000 px, 37.5 % of the image).

Working on the ROI reduces the computational cost and, above all, removes structures that are not of interest but have the same intensity as the target. As a quick check, we applied the threshold sweep of Section 2 to the whole images: the best Dice was only 0.007 for the lesion, because any threshold that selects it also selects the black background, the CSF and the sulci, and 0.561 for the lung, because the left lung has the same intensity as the right one.

Reference segmentations. To evaluate the methods quantitatively, both structures were segmented manually in 3D Slicer (Segment Editor) on the full images. The lesion reference has 280 px (3.7 % of the ROI) and the lung reference 990,686 px (56.0 % of the ROI). Slicer exported the lung segment cropped to its own bounding box, so the mask was placed back in the image grid with the origin and directions stored in the .nrrd header, and then both references were cropped with the same ROI as the images and saved as binary masks.

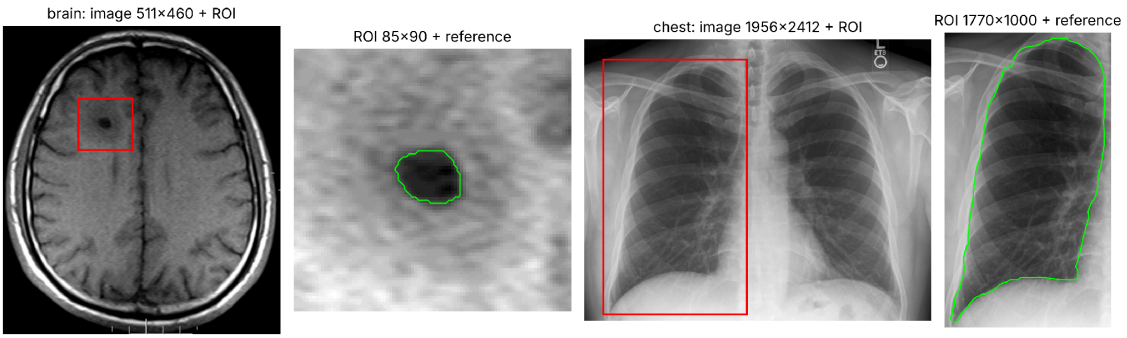


Fig 1 : Full images with the ROI (red) and cropped ROIs with the manual reference from 3D Slicer (green)

Metrics and choice of parameters. Every segmentation was compared with the reference using the Dice Similarity Coefficient, Dice = 2|A ∩ B| / (|A| + |B|), which measures the overlap (1 = identical); the Hausdorff distance (in px), which is the largest distance between the two contours and detects leaks far from the border; and the SSIM between the two binary masks. For each method, its adjustable parameters were swept and the value with the highest Dice was kept. This gives the best result that each method can reach, so it is an optimistic estimate: the parameters are tuned with the same reference used to evaluate them.

2. Thresholding

Thresholding separates the pixels of the image only by their grey level. In both ROIs the structure is darker than its surroundings (the lesion is hypointense in T1 and the aerated lung attenuates few X-rays), so the mask is formed by the pixels with I ≤ T. **Otsu's method** chooses T automatically by

maximising the variance between the two classes, which works well when the histogram is bimodal and both classes have a similar size. We compared Otsu with the best threshold of a sweep of 100 values of T between the 1st and 99th percentiles of the ROI intensities.

Brain MRI. Otsu gave T = 0.309 and a Dice of only 0.149. The lesion is only 3.7 % of the ROI, so the histogram is dominated by the white and grey matter, and Otsu separates these two tissues instead of the lesion. The best threshold, T = 0.143, lies in the valley between the small lesion mode (around 0.03-0.12) and the tissue mode, and gives Dice = 0.998: 279 px, with no false positives and a single false negative. This almost perfect result shows that the boundary drawn in 3D Slicer follows practically the same grey-level transition. The Dice vs T curve has a narrow peak: when T increases the mask quickly leaks into the white matter, and above T ≈ 0.3 the Dice is below 0.2.

Chest X-ray. Here the histogram is broad, with a lung mode around 0.2-0.35 and a second mode of bone and soft tissue around 0.6. Otsu gave T = 0.450 (Dice = 0.908) and the best threshold was T = 0.430 (Dice = 0.911), so Otsu works well because both classes have a similar size. The curve is flat between 0.40 and 0.50, so the result is not very sensitive to T. However, the mask has 99,333 false positive pixels, mainly in the soft tissue between the clavicle and the first ribs, 78,980 false negatives at the vessels of the hilum and at the borders, and 1,734 separate connected components, because thresholding does not use any spatial information.

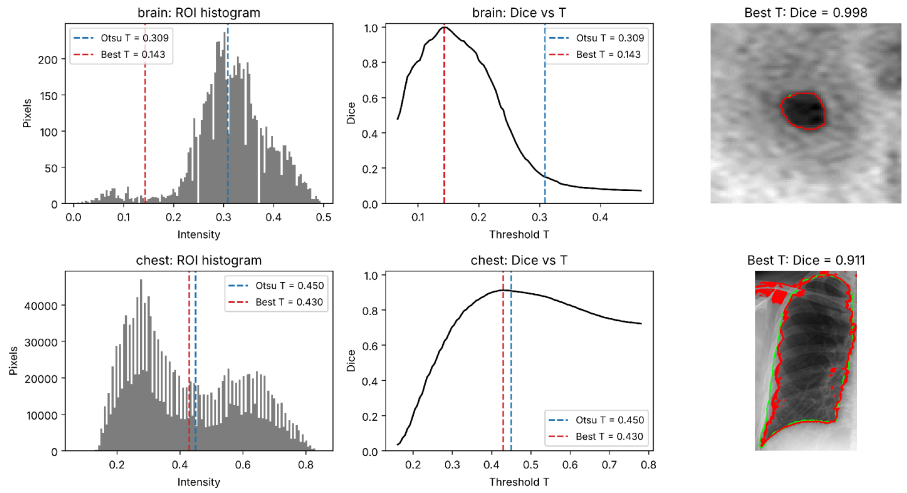


Fig 2 : ROI histograms with the Otsu and best thresholds, Dice as a function of T, and best mask (red) vs reference (green)

3. Region Growing

Region growing starts from a set of seeds placed inside the structure and adds a neighbouring pixel p to the region if $|I(p) - \text{mean(seeds)}| \leq \text{tol}$ and it is connected to the region through pixels that also meet the criterion, using 4 (sides) or 8 (sides and diagonals) neighbours. The reference value is the mean of the seeds, so the result does not depend on the order in which pixels are visited. We used one seed in the centre of the lesion, (162, 168), with intensity 0.055, and three seeds in the upper, middle and lower lung, (700, 600), (1000, 550) and (1300, 700), with a mean of 0.250. The tolerance was swept from 0.01 to 0.40 (step 0.01) with both connectivities, 80 combinations per image. To make this sweep fast on the 1.77 Mpx chest ROI, the breadth-first search was replaced by connected-component labelling, which gives exactly the same region and is much faster.

Brain MRI. The best tolerance was 0.09 with both connectivities (Dice = 0.998). With a reference of 0.055 it accepts the pixels with $I \leq 0.145$, almost the same criterion as the best threshold, and since the lesion is compact and isolated the mask is identical to the thresholding one. **Chest X-ray.** The best result was tol = 0.21 with 4-connectivity (Dice = 0.919; with 8-connectivity tol = 0.20 gives the same Dice). Compared with thresholding, the isolated spots disappear because every

pixel must be connected to a seed, and there are fewer holes at the vessels. However, the leak into the region of the clavicle remains, because it is connected to the lung apex. With small tolerances the region stops too early (Dice ≈ 0 at tol = 0.01 for the lung) and with large ones it leaks into the surrounding tissue. Both connectivities gave practically the same curves, since a leak through a single diagonal pixel did not appear in our images.

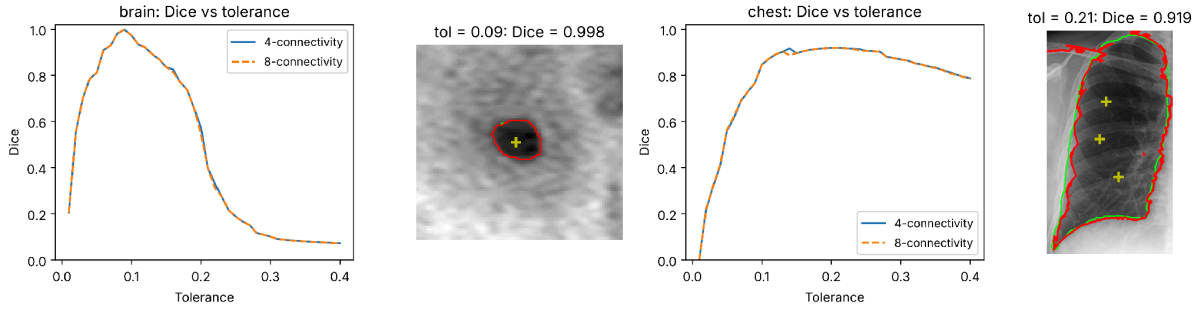


Fig 3 : Dice vs tolerance for 4- and 8-connectivity, and best region (red) with the seeds (yellow) vs reference (green)

4. Watershed

The watershed algorithm considers the image as a topographic relief and floods it from its regional minima; the lines where the floods coming from different minima meet are the region boundaries. As relief we used the gradient magnitude, so the boundaries fall on the edges. Without markers every regional minimum floods its own basin, which produces a strong over-segmentation: **696 regions** in the brain ROI and **276,629** in the chest ROI (Fig 4). Therefore we used a **marker-controlled watershed**: the object markers are small disks (radius 3 px) on the same seeds as in region growing, and the background marker is the border (frame) of the ROI. We compared two gradients, **Sobel** and **morphological** (dilation − erosion), each one after a Gaussian smoothing with σ = 0, 1, 2, 4, 8 and 12.

The best results were the Sobel gradient with σ = 4 for the lesion (Dice = 0.964; morphological with σ = 2 gave 0.957) and the morphological gradient with σ = 8 for the lung (Dice = 0.965; Sobel with σ = 12 gave 0.961). Smoothing is essential: without it (σ = 0) the noise creates many small ridges where the flood stops, and the Sobel gradient gives a Dice of about 0.92 in both images. The size of the markers had little influence as long as they stayed inside the structure, but in a test without smoothing a marker of radius 20 px, larger than the lesion, dropped its Dice to 0.17. In the lung, watershed gives a clean border without holes or isolated spots, and only a thin strip leaks into the area of the clavicle.

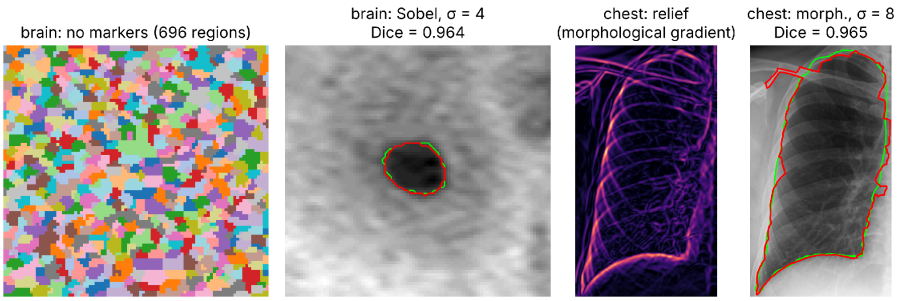


Fig 4 : Over-segmentation without markers, best marker-controlled watershed (red) vs reference (green) and morphological gradient used as relief

5. Pre- and Post-processing

Our hypothesis was that the main errors of the previous methods (isolated spots, holes at the vessels and leaks into the clavicle region) could be reduced with pre-processing, which acts on the noise and contrast of the image, and with morphological post-processing, which acts on the shape

of the mask. To test it, we applied four pre-processing methods (Gaussian filter σ = 2, median filter of radius 3, CLAHE and global histogram equalisation) and re-ran each method with its **full parameter sweep**, so that every method could adapt its parameters to the new image (Table 1).

Pre-processing	Brain: Thr.	Brain: RG	Brain: WS	Chest: Thr.	Chest: RG	Chest: WS
None (original)	**0.998**	**0.998**	0.964	0.911	0.919	**0.965**
Gaussian (σ = 2)	0.986	0.987	**0.977**	**0.912**	0.919	**0.965**
Median (r = 3)	0.986	0.986	0.969	**0.912**	0.920	**0.965**
CLAHE	0.991	0.991	0.966	0.890	**0.946**	0.964
Hist. equalisation	0.946	0.995	0.594	0.911	0.920	0.680

Table 1 : Best Dice of thresholding (Thr.), region growing (RG) and watershed (WS) after each pre-processing

For the lesion, no pre-processing improved thresholding or region growing, which were already at 0.998: the filters blur the edge of such a small object (Dice 0.986). Gaussian smoothing did help the watershed (0.964 → 0.977), because it removes small gradient ridges. For the lung, the filters hardly changed anything, while CLAHE clearly improved region growing (0.919 → 0.946) but worsened thresholding (0.911 → 0.890): CLAHE enhances the local contrast, which helps the region stop at the lung border, but it modifies the intensities differently in each zone, so a single global threshold no longer fits the whole lung. Histogram equalisation strongly degrades the watershed (0.594 and 0.680), since it amplifies small intensity differences inside homogeneous tissue and creates strong false gradients.

As post-processing, we applied to the best masks an **opening** with a disk of radius r (swept from 0 to 30 px), **hole filling** and selection of the **largest connected component**. For the lesion nothing changed (best r = 0), and an opening of r ≥ 10 erased the whole lesion, which is only about 19 px wide. For the lung, the thresholding mask improved from 0.911 to **0.934** (r = 30, the largest radius tested), since the opening removes the thin strips of the clavicle region and the isolated spots and the hole filling closes the vessels; watershed improved from 0.965 to **0.968** (r = 30); and region growing only from 0.919 to 0.920 (r = 1), because its leak is attached to the lung through a wide area and a larger radius also cuts the lung border. The large disk leaves a rounded, “bubbly” border (Fig 5), so part of the fine detail is lost in exchange for removing the leaks.

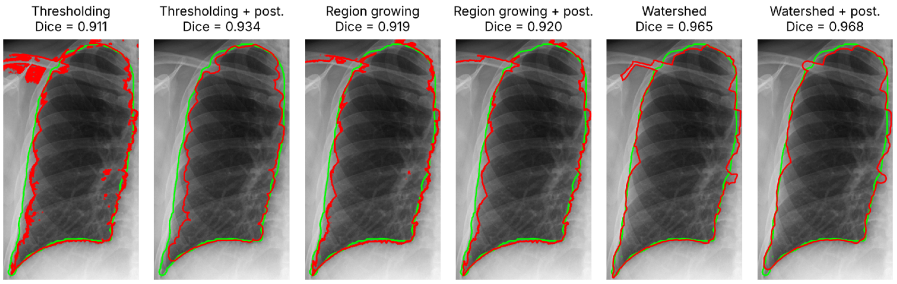


Fig 5 : Chest X-ray masks before and after post-processing (red) vs reference (green)

6. Comparison of the Part 1 Methods

Method	Brain: Dice	Brain: HD (px)	Chest: Dice	Chest: HD (px)
Thresholding	**0.998**	1.0	0.911	419.1
Thresholding + post.	**0.998**	1.0	0.934	130.4

Method	Brain: Dice	Brain: HD (px)	Chest: Dice	Chest: HD (px)
Region growing	**0.998**	1.0	0.919	419.1
Region growing + post.	**0.998**	1.0	0.920	418.6
Watershed	0.964	1.0	0.965	219.9
Watershed + post.	0.964	1.0	**0.968**	**63.2**

Table 2 : Dice and Hausdorff distance (HD) of the Part 1 methods with their best parameters

For the **lesion**, we would choose **thresholding**: it is the simplest method and reaches Dice = 0.998 with a Hausdorff distance of 1 px, because the lesion is the only structure of the ROI in its intensity range. Region growing gives the same result and would be the safer option if other dark structures appeared in the ROI, since it only keeps what is connected to the seed. For the **lung**, we would choose the **marker-controlled watershed with post-processing** (Dice = 0.968). The Hausdorff distance shows the difference better than the Dice: thresholding and region growing reach Dice ≈ 0.91-0.92, which looks acceptable, but their Hausdorff distance is 419 px because of the leak into the clavicle region, while the watershed reduces it to 63 px.

Part 2: Active Contours (Snakes)

7. Snakes

A **snake** is a closed curve v(s), defined by a set of points, that moves over the image until it minimises an energy made of an internal term, which controls the shape of the curve, and an external term, which attracts it to the features of the image:

$$E_{snake} = \int [\, \alpha |v'(s)|^2 + \beta |v''(s)|^2 - w_{line}\, I(v) - w_{edge}\, |\nabla I(v)|^2 \,]\, ds$$

Here α is the **elasticity** (it penalises stretching, so the curve shrinks), β is the **rigidity** (it penalises bending, so the curve stays smooth), w_{edge} attracts the curve to strong gradients and w_{line} to bright or dark intensities. We used the "simple" active contour of scikit-image (*active_contour*) with w_{edge} = 1, w_{line} = 0, time step γ = 0.001 and up to 2500 iterations, after a Gaussian smoothing σ that lets the edges be "felt" from further away. The snake is a local method, so it needs an initial curve around the structure: a circle of radius 16 px (100 points, σ = 1) for the lesion and an ellipse with radii 840 × 480 px (400 points, σ = 2) for the lung. For the lung the ROI was reduced 8 times before running the snake: at full resolution the edge force is too weak compared with the elasticity, and the curve does not stop at the lung border (Dice = 0.340 at scale 1, 0.849 at 2, 0.827 at 4, **0.960 at 8** and 0.940 at 16), and it is also much slower.

β \ α	0.001	0.005	0.015	0.05		0.001	0.005	0.015	0.05
0.1	0.929	0.911	0.873	0.000		0.957	0.955	0.950	0.895
1	0.935	0.918	0.932	0.932		0.956	0.951	0.950	0.894
10	0.907	**0.939**	0.913	0.907		**0.960**	0.953	0.943	0.888

Table 3 : Dice of the snake for each α (columns) and β (rows): Brain MRI (left) and Chest X-ray (right)

Influence of the parameters. The best combination was α = 0.005, β = 10 for the lesion (Dice = 0.939) and α = 0.001, β = 10 for the lung (Dice = 0.960). A high elasticity with a low rigidity (α = 0.05, β = 0.1) made the small contour collapse completely (Dice = 0), and for the lung α = 0.05 always gave the worst results (≤ 0.895), because the curve shrinks too much. A high rigidity (β = 10) was the best option in both images, since it keeps the curve smooth and stops it from entering small irregularities. The smoothing must be adapted to the size of the structure: for the lesion the

Dice was 0.938 with σ = 0.5, 0.939 with σ = 1, 0.908 with σ = 2, 0.822 with σ = 3 and 0 with σ = 4, because a ~19 px lesion is completely blurred.

Sensitivity to the initial contour. Keeping the best parameters, we changed only the initial curve (Fig 6). For the lesion, a tighter (r = 10) or larger (r = 25) circle still gave Dice = 0.916 and 0.934, but a circle shifted 12 px to the right, partly outside the lesion, collapsed to Dice = 0. For the lung, a smaller ellipse gave 0.815, an ellipse shifted 130 px gave 0.879, and an ellipse placed inside the lung only 0.412, because the snake can only shrink towards the nearest edges and cannot grow to reach the real border.

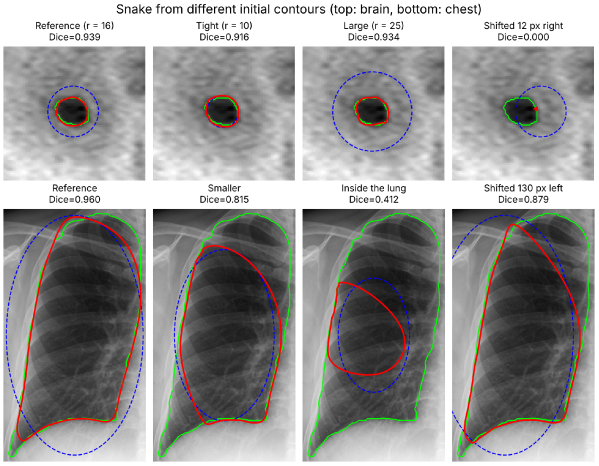


Fig 6 : Snake from different initial contours (blue dashed): final snake (red) vs reference (green). Top: Brain MRI, bottom: Chest X-ray

Comparison with Part 1. For the lesion the snake (Dice = 0.939, HD = 1.4 px, SSIM = 0.971) is worse than thresholding (0.998), because its smooth contour cannot follow the small irregularities of the lesion border. For the lung the snake reaches Dice = 0.960 and HD = 137.4 px, very close to the watershed (Dice = 0.965, and 0.968 with HD = 63.2 px after post-processing) and much better than thresholding and region growing in terms of Hausdorff distance (419 px): as the curve is closed and smooth, it does not leak into the clavicle region, although it cuts the lung apex and the costophrenic angle, which have sharp corners.

Part 3: Other Segmentation Methods

8. Chan-Vese Segmentation

In the state of the art there are three main families of more advanced segmentation methods: **region-based active contours and level sets**, such as Chan-Vese; **graph-based methods**, such as graph cuts or the random walker; and **deep learning**, with networks such as U-Net or foundation models such as MedSAM, which give the best results but need large annotated datasets or pre-trained models. We selected **Chan-Vese** (*active contours without edges*) because it is the natural continuation of the snake of Part 2, but it removes its main weakness: it does not depend on the edges, and it needs neither training data nor seeds. The image is approximated by two regions of constant intensity, c_1 inside the curve C and c_2 outside, and the method minimises:

$$F(c_1, c_2, C) = \mu \cdot Length(C) + \nu \cdot Area(inside\ C) + \lambda_1 \int_{inside} |I - c_1|^2\, dx + \lambda_2 \int_{outside} |I - c_2|^2\, dx$$

The curve stops where the **mean intensity** changes, not where the gradient is high. It is the zero level of a level set function φ, so it can split and merge, which a snake cannot do. μ controls the smoothness of the contour and λ_1, λ_2 weight the fit inside and outside ($\lambda_1 > \lambda_2$ forces the object to be more homogeneous); ν is 0 in scikit-image. We used *skimage.segmentation.chan_vese* (Getreuer's implementation, up to 500 iterations), starting from the **same initial ellipse and**

downsampling as the snake, and kept as object the phase that contains the centre of the ellipse. With the default parameters (μ = 0.25, $\lambda_1 = \lambda_2$ = 1), this initialisation gave Dice = 0.753 for the lesion and 0.903 for the lung, while the default checkerboard initialisation gave 0.136 and 0.914.

Influence of the parameters. We swept $\mu \in \{0.05, 0.1, 0.25, 0.5\}$ and $\lambda_1 \in \{1, 2, 5, 10\}$ with λ_2 = 1. For the lesion, increasing λ_1 was decisive: it forces the inside to contain only the darkest pixels and removes the grey halo around the lesion, and the best result was μ = 0.05, λ_1 = 10 with Dice = **0.993**. For the lung the best was μ = 0.5, λ_1 = 2 (Dice = **0.917**), while λ_1 = 10 shrank the lung to its darkest part (Dice 0.74-0.77). In the lung, Chan-Vese also includes the soft tissue above the clavicle, which has a mean intensity similar to the lung.

Robustness. First, Chan-Vese was run from the same initial curves as the snake (Fig 7). It handles small or interior curves better than the snake, because the level set can also grow: 0.827 vs 0.815 with a smaller ellipse and 0.610 vs 0.412 with an ellipse inside the lung. However, for the lesion a large circle (r = 25) gave only 0.075 (snake: 0.934) and the shifted circle 0.071, partly because the object phase is chosen by the centre of the ellipse, which then falls outside the lesion. So, with the tuned λ_1 = 10, Chan-Vese depends more on the initialisation than expected. Second, we added Gaussian noise of increasing σ to the ROIs and ran all the methods with their clean-image parameters (Fig 8). For the lesion Chan-Vese fell from 0.993 to 0.524 at σ = 0.2, since its low μ accepts many small fragments, while watershed (0.852) and the snake (0.741) were the most robust and thresholding the least (0.192). For the lung, Chan-Vese kept 0.917 at every noise level, mainly because the 8× downsampling averages the noise; watershed (0.958) and the snake (0.956) were also stable, while thresholding and region growing dropped to 0.750 and 0.732.

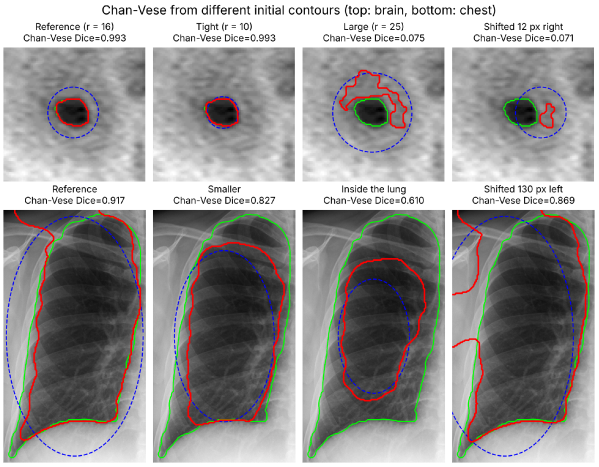


Fig 7 : Chan-Vese from the same initial contours as the snake (blue dashed): result (red) vs reference (green)

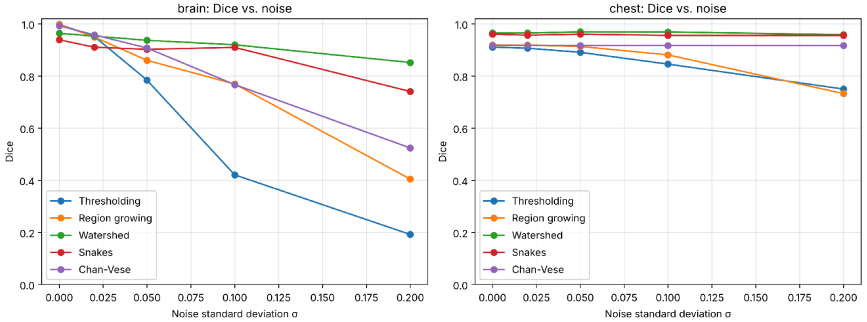


Fig 8 : Dice of every method as a function of the standard deviation of the added Gaussian noise

9. Final Comparison and Conclusions

Method	Brain: Dice	Brain: HD (px)	Brain: SSIM	Chest: Dice	Chest: HD (px)	Chest: SSIM
Thresholding	**0.998**	1.0	**0.999**	0.911	419.1	0.856
Thresholding + post.	**0.998**	1.0	**0.999**	0.934	130.4	0.911
Region growing	**0.998**	1.0	**0.999**	0.919	419.1	0.880
Watershed + post.	0.964	1.0	0.982	**0.968**	**63.2**	**0.946**
Snakes	0.939	1.4	0.971	0.960	137.4	0.940
Chan-Vese	0.993	1.0	0.996	0.917	354.1	0.891

Table 4 : Final comparison of the main methods of Parts 1, 2 and 3 against the manual reference

For the **brain lesion**, a small and isolated structure with a clear intensity difference, the simplest methods were the best: thresholding and region growing reached Dice = 0.998, followed by Chan-Vese (0.993), which was very close but more sensitive to the initial curve and to noise. For the **right lung**, a large structure surrounded by tissue of similar intensity, the methods that use edges together with markers or a shape constraint were clearly better: watershed with post-processing obtained the best Dice (0.968) and the lowest Hausdorff distance (63 px), followed by the snake (0.960), while thresholding, region growing and Chan-Vese leaked into the clavicle region. Finally, all the results were obtained with parameters tuned against our own manual reference, so they are an upper bound of what each method would achieve on a new image.